The Journal of Open Source Software

r4lineups: Statistical Inference on Lineup Fairness

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DOI: 10.xxxxxx/draft

Software

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Editor:

Submitted: 02 November 2026
Published: unpublished

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Summary

In eyewitness lineups, a police officer asks a witness to decide whether a person seen during an event is present among a set of alternatives, and to identify that person if present. The evidential value of a lineup depends not only on whether a suspect is chosen, but also on whether the suspect stands out, whether the fillers are plausible alternatives, and how accuracy changes with confidence and response time. These questions have generated distinct traditions of lineup-fairness measurement, confidence-based analysis, and models of lineup decisions (Malpass, 1981; Mickes et al., 2012; Tredoux, 1998; Wells et al., 1979; Wells & Lindsay, 1980).

`r4lineups` is an R package that brings those traditions into one collection. It provides classical and modern measures of lineup fairness, bootstrap and Bayesian uncertainty summaries, ROC/CAC/RAC and full-ROC analyses, calibration and evidential measures, signal-detection and two-high-threshold models, the Multi-d' framework, simulation and power tools, and optional face-similarity and lineup-memory-model workflows. An interactive application covers most of these analyses and runs in a web browser without installation. The intended users are eyewitness researchers, forensic psychologists, students, and applied analysts.

Statement of need

Mock-witness assessments and eyewitness experiments produce different but related data. Mock witnesses who did not see the perpetrator choose the lineup member who best matches a description; uneven choices indicate that some members are more plausible than others. Both lineup bias and lineup effective size are traditionally computed from mock-witness choices. Eyewitness experiments instead compare target-present and target-absent decisions, often with confidence and response time, although eyewitness data can also support estimates of lineup fairness (Fitzgerald et al., 2023). Lee and Penrod, for example, proposed the difference between the z-transformed rates at which witnesses choose the innocent suspect and the fillers from target-absent lineups as a resultant index of lineup bias (Lee & Penrod, 2019). Analysts therefore need both lineup-construction measures—including suspect-selection proportion, foil bias, functional size, Malpass's effective size, and Tredoux's effective size—and measures of identification performance and evidential value (Doob & Kirshenbaum, 1973; Malpass, 1981; Tredoux, 1998; Wells et al., 1979).

Before `r4lineups`, these computations were commonly distributed across article supplements, small bespoke scripts, and general statistical packages. That fragmentation made input conventions, denominators, base-rate assumptions, continuity corrections, and uncertainty procedures difficult to audit consistently. `r4lineups` supplies lineup-specific validation and common data conventions together with worked examples. Version 2.3.0 is a major extension of the original package, first released on CRAN as version 0.1.1 in 2018 (Tredoux & Naylor, 2026).

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State of the field

General R tools such as `boot` (Canty & Ripley, 2025) provide important computational infrastructure, but do not encode eyewitness response categories or estimands. The R package `fullROC` focuses on full receiver operating characteristic curves using suspect, filler, and rejection responses (Yang & Smith, 2023). The Python package `pyWitness` provides a broad eyewitness-analysis toolkit, including confidence-based signal-detection models (Mickes et al., 2024).

`r4lineups` occupies a complementary position. It combines mock-witness fairness inference, confidence-based evidential analysis, Bayesian summaries, model-based analyses, simulation, and reporting in an R-native interface. It does not duplicate the established Python likelihood/model-fitting engine. Instead, its optional `fit_lineup_models()` bridge calls a pinned pyWitness revision and returns R-native objects. Independent Observations, Ensemble, and Integration models can therefore be examined from R while remaining comparable with the reference implementation (Wixted et al., 2018). A separate function, `fit_max_sdt()`, fits a narrower model: a single-criterion, aggregate, equal-variance version of the Independent Observations model, not the complete confidence-based Wixted model.

Software design

The core design separates two input families. Fairness functions accept "lineup choice vectors" or "lineup tables". Confidence-based functions use trial-level data with `target_present`, `identification`, and `confidence`, with optional fields such as response time and experimental condition. Analysis wrappers return structured S3 objects with `print()`, `summary()`, and `plot()` methods, making assumptions and derived quantities inspectable while supporting normal R workflows. Two helpers make data coding explicit: `lineup_table()` tabulates choices over every lineup position, so that members nobody chose remain visible, and `classify_lineup_choices()` derives suspect, filler, and rejection responses from the chosen position and the suspect's position.

The package favours R-native calculations for its statistical core and makes external systems explicit. Face embeddings and the pyWitness bridge require separately managed Python environments; neither is initialised or installed when the package loads or during CRAN checks. Results from the lineup-model bridge contain extracted R tables rather than live Python objects, so saved analyses remain readable without an active Python session. This design trades some installation convenience for reproducibility and a clear boundary between package code and external models.

We built the interactive application from modules that call only exported package functions, so that every number it reports can be reproduced in R. Confidence-based data are loaded once, with the nominal lineup size and any grouping variable used for comparisons, and bootstrap and posterior analyses run on request with a user-supplied seed. The application is installed with the package and also runs entirely in the browser through WebAssembly, from the package website. In the browser version, mock-witness and experimental data stay on the user's computer, an important safeguard when the data come from police casework.

There are sixteen vignettes, and these provide the methodological detail that users will need. They cover fairness and diagnosticity, Bayesian inference, ROC/CAC/RAC and full ROC, calibration and decision measures, expected information gain and error-rate ranges, Winter's two-high-threshold model, summary and regression-based signal-detection methods, the Multi-d', mSDT and Wixted models, simulation and power, and optional face similarity. Validation of selected methods combines published definitions, independent hand calculations, invariance identities, numerical integration and simulation recovery. The lineup-memory bridge also checks its interface against a pinned pyWitness revision (Mickes et al., 2024). A statistical audit by the authors, assisted by OpenAI Codex, records the conditions examined, evidence gaps

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and limitations of approximate inference; selected counterexamples are retained as regression tests. The pyWitness bridge applies a per-fit correction to the pinned engine's rejection-cell variance denominators so its default objective is consistently Pearson chi-squared, without modifying the installed Python environment; an option reproduces the engine's own objective for comparison with published fits.

Research impact statement

`r4lineups` has been publicly available through CRAN since 2018 and has been used in published eyewitness research, including analyses of eyewitness decision processes (Mansour et al., 2024, 2025). Version 2.3.0 substantially extends that established research tool while preserving the classical fairness functions used by earlier work. Its immediate scholarly contribution is a tested R workflow spanning lineup construction, identification outcomes, confidence, evidence, and competing process models. The package website, sixteen vignettes, example datasets, and the browser-based application also support teaching, method comparison, and applied review.

AI usage disclosure

Generative AI tools were used in developing the software, its documentation and this paper: Anthropic Claude (Sonnet 4 and 5; Opus 4.6, 4.8 and 5.5) and OpenAI Codex with GPT-5-family models (including GPT-5.5 and GPT-5.6 Sol) and GPT-6.1 Sol. In the software, they generated and refactored code, including the interactive application, scaffolded tests, and assisted with code review, debugging and package-check remediation. They were used to assist with drafting documentation and vignettes. They helped draft this paper under close control by the authors, and checked its language for spelling and grammatical issues. The statistical audit described above was conducted by the authors with assistance from OpenAI Codex (GPT-6.1 Sol).

The authors made the statistical and software-design decisions. Colin Tredoux thoroughly reviewed, modified and validated all AI-generated content. Tamsyn Naylor programmed much of the original package and has maintained it. The authors retain full responsibility for accuracy, originality, licensing and scholarly claims.

Acknowledgements

We gratefully acknowledge the many researchers whose publications provide the computational foundations implemented in `r4lineups`, including work on lineup fairness, confidence-based analysis, evidential measures, and models of lineup memory (Fitzgerald et al., 2023; Smith et al., 2020; Starns et al., 2023; Winter et al., 2022; Wixted et al., 2018). Errors in translating that published work into package code remain the responsibility of the package authors and may be reported through the GitHub issue tracker.

This work was supported in part by incentive grants from the National Research Foundation of South Africa awarded to Colin G. Tredoux during 2016–2026 (maximum value approximately US$2,000). The funder had no role in the software's design, development, validation, manuscript preparation, or the decision to submit. The authors declare no conflicts of interest.

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